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Seyed Mohammadreza Mousavi

Education

- **Sadjad University**, Mashhad Iran
M.Sc. in Artificial Intelligence, 2017-2020
Supervisor: [Dr. Amir Bavafa Toosi](#)
Thesis: Development of English handwritten recognition using Deep Neural Network
Thesis grade: 18.5/20; Ranked in top 5% of class
- **Azad University of Mashhad**, Mashhad, Iran
B.Sc in Computer Engineering - Hardware Engineering, 2012-2016
Supervisor: [Dr. Farzaneh Kimiaei](#)
Thesis: Developing AVR programming for automation
Thesis grade: 20/20; Ranked first in class

Publications

Journal Papers:

1. **Mousavi, S. M.**, & Toosi, A. B. (2024). Recognition of characters using a gated convolutional and recurrent neural network architecture. Journal of Soft Computing and Information Technology, 13(3), 31–45.
Received: 05/08/2024; Revised: 12/12/2024; Accepted: 12/23/2024.
Available at: http://www.jscit.nit.ac.ir/article_212386.html?lang=en
2. **Mousavi, S. M.**, & Toosi, A. B. (2025). GCRCR: Gated Convolutional and Recurrent Character Recognition.
Manuscript submitted for publication in International Journal on Document Analysis and Recognition (IJDAR).
3. Hossaini, S. B., **Mousavi, S. M.**, & Toosi, A. B. (Year). A hybrid method for generating markers to improve the image segmentation of the watershed algorithm.
Manuscript submitted for publication in Tabriz Journal of Electrical Engineering (TJEE).

Conference Paper:

1. **Mousavi, S. M.**, Rahimian Zariv, A., & Hamidzadeh, J. (Year). Lithium Battery SOH Estimation: A Novel Octave and LSTM Hybrid Neural Network Approach.
Paper accepted in the 12th Iranian Conference on Renewable Energies and Distributed Generation (not yet published).
Conference website: <https://icredg2025.qut.ac.ir/>

Note: Abstracts will be provided upon request.

Research Interest

Image Processing	Deep Learning Models	Reinforcement Learning
Computer Vision	Object Class Recognition	Natural Language Processing
Machine Learning	Robotics navigation and SLAM	

Teaching Experience

Python programming language

- Teaching Python programming language to undergraduates at Industrial Sadjad University since Jan 2024.

Internet of Things & Hardware Programming

- Taught IoT with Arduino/Raspberry Pi, covering sensor integration and embedded system design.

Honors and Awards

- **M.Sc. Admission with distinction (Spring 2019)**
Admitted to the Master's program in AI at Industrial Sadjad University as a top student.
- **Exclusive Teaching Invitation**
Selected as an instructor at Industrial Sadjad University while pursuing M.Sc., chosen among peers.
- **Outstanding Faculty Recognition (2024)**
Recognized by Sadjad Industrial University for excellence in research and teaching:
https://www.linkedin.com/in/seyed-mohammadreza-mousavi/overlay/1742026180876/single-media-viewer/?profileId=ACoAACxGXmkBPzeK1VQ_zy6_sEAKJVPpiqAoxJ0

Skills

Programming languages

Python, Mathlab, C, C++, PHP, R, Arduino, Flutter

Technical tools

OpenMMLab, PyTorch, Tensorflow, Keras, ROS, Scikit Learn, Matplotlib, Pandas, SciPy, Numpy

Other skills

Familiar with: Linux (Lpic 1&2), Version Control (Git&Github), Docker

Working Experience

Software-Motion, China, Suzhou - *Calibration algorithm Engineer (remote)*

Jul 2024 – Now

Supervisor: 叶辰怡 (Tiffany)

- Built **ROS/SLAM-based AGV** software with sensor fusion and navigation.
- Enabled reliable localization in industrial and outdoor settings.

Perception model team head, China, Suzhou

Oct 2023 – Jun 2024

- Optimized traffic light detection using **anchor-based/anchor-free** models (RetinaNet, YOLOX).
- Built ONNX conversion pipelines for cross-platform model deployment.

VERA startup, Mashhad, Iran - *AI developer*

Aug 2023 - Oct 2023

- Built and deployed an **NLP-based conversational AI** with persistent user context.
- Led development of a TTS model to improve audio interactions.

FAV Information Technology, Mashhad, Iran - *R & D Engineer*

Aug 2021 - 2023

- Built ML models in Python to forecast market trends and segment customers.
- Optimized inventory levels using demand pattern analysis to reduce costs.

Samatoos, Mashhad, Iran - *Network and Computer SysAdmin*

May 2016 - 2021

- Managed VMs with VMware vSphere and Hyper-V; implemented backup and recovery solutions.

- Deployed Zabbix for proactive network monitoring and issue resolution.

Test Score

- IELTS overall: 6.5 (L=6, R=6.5, W=6, S=6.5)

Projects

Retinal Blood Vessel Segmentation

Jul 2023

Built a RetinaNet blood vessel segmentation model in TensorFlow with pre-trained encoder: https://github.com/seyed-mohammadreza-mousavi/Retinal-Vessel-Segmentation_A-Computer-Vision-Technique

Object Detection with YOLOv3

May 2023

Implemented a real-time Object Detection system with YOLOv3, fine-tuning parameters and evaluating performance using mAP.

Overtime Work Forecasting with Neural Prophet

Mar 2022

Developed a customized Neural Prophet forecasting system to predict staff overtime, analyzing historical data for key patterns.

Handling Imbalanced Datasets with Chaotic Functions and Evolutionary algorithms

Jan 2022

Designed a chaotic-evolutionary method to balance datasets with synthetic samples.

- Source code available upon request.

AI passed courses

Reinforcement Learning, Evolutionary Algorithms, Machine Vision, Pattern Recognition, Machine Learning, Artificial Intelligence, Fuzzy system.

Online courses and degrees

Introduction to Self-Driving Cars ([Coursera](#))

ROS for Beginners: Basics, Motion, and OpenCV ([Udemy](#))

Supervised Machine Learning: Regression and Classification ([Coursera](#))

Advance Your Skills in Deep Learning and Neural Networks ([LinkedIn](#))

Deep Learning: Image Recognition ([LinkedIn](#))

Machine Learning with Python: Foundations ([LinkedIn](#))

References

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